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Propiedad de la convolución para exponentes conjugados

Proposición 1. *Sean $f \in \mathcal{L}^p(\mathbb{R}^n)$, $g \in \mathcal{L}^q(\mathbb{R}^n)$ con $p, q \in [1, \infty]$ conjugados, entonces*

(1) $f * g \in \mathcal{L}^\infty(\mathbb{R}^n)$ *y* $\|f * g\|_\infty \leqslant \|f\|_p \|g\|_q$.

(2) $f * g$ *es uniformemente continua en* $\mathbb{R}^n$.

***Demostración*:**

(1) Aplicando la desigualdad de Hölder a la definición de convolución, tenemos que

$$|f * g(x)| = \left| \int_{\mathbb{R}^n} f(x-y)\, g(y)\, \mathrm{d}y \right| \leqslant \int_{\mathbb{R}^n} |f(x-y)|\, |g(y)|\, \mathrm{d}y \leqslant \|f\|_p \|g\|_q .$$

Por tanto, $\|f * g\|_\infty \leqslant \|f\|_p \|g\|_q$.

(2) Definimos $\tilde{f}(x) := f(-x)$ y consideramos $p \in [1, \infty)$. Por el `lem-convergencia-lp-traslacion/Ler` dado $\varepsilon > 0$, existe $\delta > 0 : \forall h \in B(0, \delta) : \left\| T_h \tilde{f} - \tilde{f} \right\|_p < \frac{\varepsilon}{\|g\|_q}$. Sean $a, b \in \mathbb{R}^n$ tales que $\|a - b\| < \delta$, entonces

$$\begin{aligned}
|f * g(b) - f * g(a)| &= \left| \int_{\mathbb{R}^n} \left(f(b-y) - f(a-y)\right) g(y)\, \mathrm{d}y \right| \\
&\leqslant \int_{\mathbb{R}^n} \left| T_b \tilde{f}(y) - T_a \tilde{f}(y) \right| |g(y)|\, \mathrm{d}y \\
&\overset{\substack{\text{Hölder} \\ \downarrow}}{\leqslant} \left(\int_{\mathbb{R}^n} \left| T_b \tilde{f}(y) - T_a \tilde{f}(y) \right|^p \mathrm{d}y \right)^{\frac{1}{p}} \|g\|_q \\
&= \left(\int_{\mathbb{R}^n} \left| T_a \left(T_{b-a} \tilde{f} - \tilde{f} \right)(y) \right|^p \mathrm{d}y \right)^{\frac{1}{p}} \|g\|_q \leqslant \frac{\varepsilon}{\|g\|_q} \|g\|_q = \varepsilon.
\end{aligned}$$

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